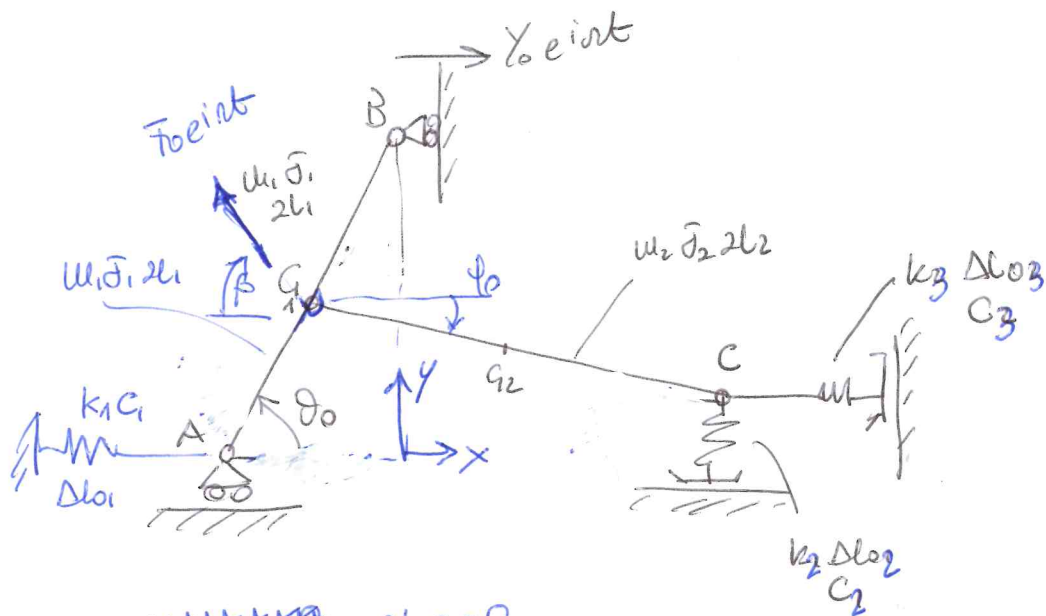




$$\underline{x} = \begin{Bmatrix} \theta \\ \varphi \\ y \end{Bmatrix}$$

$$\underline{x}_0 = \begin{Bmatrix} \theta_0 \\ \varphi_0 \\ 0 \end{Bmatrix}$$

$$x_A = y - 2L_1 \cos \theta$$

$$x_{G1} = y - L_1 \cos \theta$$

$$y_{G1} = L_1 \sin \theta$$

$$x_{G2} = y - L_1 \cos \theta + L_2 \cos \varphi$$

$$y_{G2} = L_1 \sin \theta - L_2 \sin \varphi$$

$$x_C = y - L_1 \cos \theta + 2L_2 \cos \varphi$$

$$y_C = L_1 \sin \theta - 2L_2 \sin \varphi$$

$$\Delta l_1 = x_A - x_{Ap}$$

$$\Delta l_2 = y_C - y_{Cp}$$

$$\Delta l_3 = -x_C + x_{Cp}$$

$$\frac{\partial x_{G1}}{\partial \theta} = L_1 \sin \theta \quad \frac{\partial x_{G1}}{\partial \varphi} = 0 \quad \frac{\partial x_{G1}}{\partial y} = 1$$

$$\frac{\partial y_{G1}}{\partial \theta} = L_1 \cos \theta \quad \frac{\partial y_{G1}}{\partial \varphi} = 0 \quad \frac{\partial y_{G1}}{\partial y} = 0$$

$$\frac{\partial x_{G2}}{\partial \theta} = L_1 \sin \theta \quad \frac{\partial x_{G2}}{\partial \varphi} = -L_2 \sin \varphi \quad \frac{\partial x_{G2}}{\partial y} = 1$$

$$\frac{\partial y_{G2}}{\partial \theta} = L_1 \cos \theta \quad \frac{\partial y_{G2}}{\partial \varphi} = -L_2 \cos \varphi \quad \frac{\partial y_{G2}}{\partial y} = 0$$

$$\frac{\partial \Delta l_1}{\partial \theta} = -L_1 \sin \theta \quad \frac{\partial \Delta l_1}{\partial \varphi} = 0 \quad \frac{\partial \Delta l_1}{\partial y} = 1$$

$$\frac{\partial \Delta l_2}{\partial \theta} = L_1 \cos \theta \quad \frac{\partial \Delta l_2}{\partial \varphi} = -L_2 \sin \varphi \quad \frac{\partial \Delta l_2}{\partial y} = 0$$

$$\frac{\partial \Delta l_3}{\partial \theta} = -L_1 \sin \theta \quad \frac{\partial \Delta l_3}{\partial \varphi} = +2L_2 \sin \varphi \quad \frac{\partial \Delta l_3}{\partial y} = -1$$

$$\Delta u = \begin{bmatrix} L_1 \sin \theta_0 & 0 & 1 \\ L_1 \cos \theta_0 & 0 & 0 \\ 1 & 0 & 0 \\ L_1 \sin \theta_0 & -L_2 \sin \phi_0 & 1 \\ L_1 \cos \theta_0 & -L_2 \cos \phi_0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Delta h = \Delta c = \begin{bmatrix} 2L_1 \sin \theta_0 & 0 & 1 \\ L_1 \cos \theta_0 & -2L_2 \cos \phi_0 & 0 \\ -L_1 \sin \theta_0 & 2L_2 \sin \phi_0 & -1 \end{bmatrix}$$

$$\delta W = -F \cos \beta \delta x_{g1} + F \sin \beta \delta y_{g1}$$

$$\vec{F} = \begin{Bmatrix} -\cos \beta \\ \sin \beta \end{Bmatrix} F_0 \text{ einb}$$

$$\Delta q = \begin{bmatrix} L_1 \sin \theta_0 & 0 & 1 \\ L_1 \cos \theta_0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta u_1}{\partial \theta^2} = 2L_1 \cos \theta \quad \frac{\partial^2 \Delta u_1}{\partial \phi^2} = 0$$

$$[K_{E, \Pi, 1}] = k_1 \Delta l_1 \begin{bmatrix} 2L_1 \cos \theta_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta u_2}{\partial \theta^2} = -L_1 \sin \theta \quad \frac{\partial^2 \Delta u_2}{\partial \phi^2} = 2L_2 \sin \phi$$

$$[K_{E, \Pi, 2}] = k_2 \Delta l_2 \begin{bmatrix} -L_1 \sin \theta_0 & 0 & 0 \\ 0 & 2L_2 \sin \phi_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

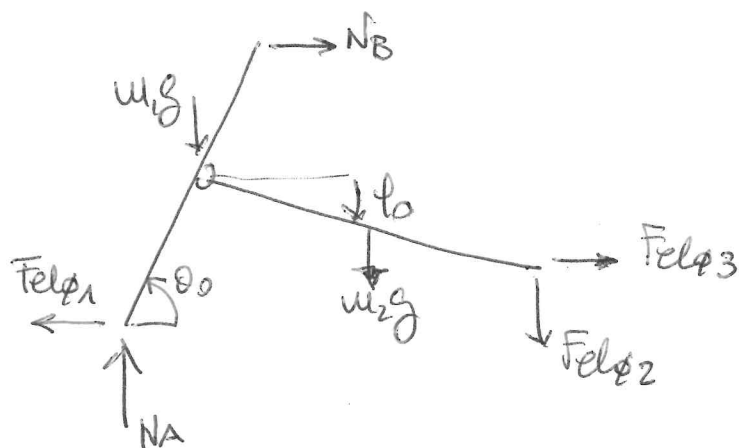
$$\frac{\partial^2 \Delta u_3}{\partial \theta^2} = -L_1 \cos \theta \quad \frac{\partial^2 \Delta u_3}{\partial \phi^2} = 2L_2 \cos \phi$$

$$[K_{E, \Pi, 3}] = k_3 \Delta l_3 \begin{bmatrix} -L_1 \cos \theta_0 & 0 & 0 \\ 0 & 2L_2 \cos \phi_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 h_{s1}}{\partial \theta^2} = -L_1 \sin \theta \quad \frac{\partial^2 h_{s1}}{\partial \phi^2} = 0$$

$$\frac{\partial^2 h_{s2}}{\partial \theta^2} = -L_1 \sin \theta \quad \frac{\partial^2 h_{s2}}{\partial \phi^2} = L_2 \sin \phi$$

$$[K_{g1}] = m_{g1} \begin{bmatrix} -L_1 \sin \theta_0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad [K_{s1}] = m_{g1} \begin{bmatrix} -L_1 \sin \theta_0 & 0 & 0 \\ 0 & L_2 \sin \phi_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



$$\sum F_x = 0 : F_{elphi_3} + N_B - F_{elphi_1} = 0$$

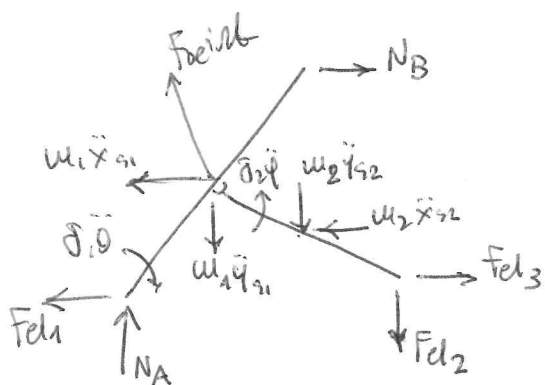
$$\sum F_y = 0 : N_A - w_1 g - w_2 g - F_{elphi_2} = 0$$

$$\sum M_{AB} = 0 : N_B L_1 \sin \theta_0 + F_{elphi_1} L_1 \sin \theta_0 + N_A L_1 \cos \theta_0 = 0$$

$$\sum M_{B1} = 0 : F_{elphi_2} 2L_2 \cos \phi_0 - F_{elphi_3} 2L_2 \sin \phi_0 + w_2 g L_2 \cos \phi_0 = 0$$

$$\underline{x} = \{ F_{elphi_2} \quad F_{elphi_3} \quad N_A \quad N_B \}^T$$

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & L_1 \cos \theta_0 & L_1 \sin \theta_0 \\ 2L_2 \cos \phi_0 & -2L_2 \sin \phi_0 & 0 & 0 \end{bmatrix} \underline{x} = \begin{Bmatrix} F_{elphi_1} \\ (w_1 + w_2)g \\ -F_{elphi_1} L_1 \sin \theta_0 \\ -w_2 g L_2 \cos \phi_0 \end{Bmatrix}$$



$$N_B = F_y = (-M_{cf} \ddot{\theta} + i C_{cf} \dot{\theta} + k_{cf}) \underline{x}_F + (-k_{cc} \ddot{\theta} + i c_{cc} \dot{\theta} + k_{cc}) \underline{y}_0$$

$$N_B = w_1 \ddot{x}_{s1} + w_2 \ddot{x}_{s2} + F_0 \cos \beta - F_{elphi_3} + F_{elphi_1}$$

$$N_A = w_1 \ddot{y}_{s1} + w_2 \ddot{y}_{s2} + F_{elphi_2} - F_0 \sin \beta$$